

Midterm

$$\frac{d}{dt}({}^{\mathcal{B}}\mathbf{r})_{\mathcal{N}} = \frac{d}{dt}({}^{\mathcal{B}}\mathbf{r})_{\mathcal{B}} + {}^{\mathcal{B}}\boldsymbol{\omega}_{\mathcal{B}/\mathcal{N}} \times {}^{\mathcal{B}}\mathbf{r} \quad \mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$$

$${}^{\mathcal{B}}\boldsymbol{\omega}_{\mathcal{B}/\mathcal{N}} = {}^{\mathcal{B}}\boldsymbol{\omega}_{\mathcal{B}} - {}^{\mathcal{B}}\boldsymbol{\omega}_{\mathcal{N}} \quad \boldsymbol{\omega} = \frac{\mathbf{r} \times \mathbf{v}}{r^2}$$

$$\begin{bmatrix} {}^{\mathcal{N}}\mathbf{r}_{\mathcal{B}} \\ {}^{\mathcal{N}}\mathbf{v}_{\mathcal{B}} \end{bmatrix} = \begin{bmatrix} {}^{\mathcal{N}}\mathbf{R} & {}^{\mathcal{N}}\mathbf{R} \\ {}^{\mathcal{B}}\boldsymbol{\omega}_{\mathcal{B}/\mathcal{N}} \times \end{bmatrix} \begin{bmatrix} {}^{\mathcal{B}}\mathbf{r}_{\mathcal{B}} \\ {}^{\mathcal{B}}\mathbf{v}_{\mathcal{B}} \end{bmatrix} \quad {}^{\mathcal{B}}\boldsymbol{\omega}_{\mathcal{B}/\mathcal{N}} = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

$$\begin{bmatrix} {}^{\mathcal{B}}\boldsymbol{\omega}_{\mathcal{B}/\mathcal{N}} \times \end{bmatrix} = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \iff {}^{\mathcal{B}}\boldsymbol{\omega}_{\mathcal{B}/\mathcal{N}} = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

$$r = \frac{p}{1 + e \cos f} \quad \dot{r} = \frac{r^2 \dot{f} e \sin f}{p} \quad p = a(1 - e^2) = \frac{h^2}{\mu}$$

$$a = \frac{r_p + r_a}{2} \quad e = \frac{r_a - r_p}{r_a + r_p} \quad b = a\sqrt{1 - e^2}$$

$$n = \frac{h}{a^2\sqrt{1 - e^2}} = \frac{2\pi}{T} \quad M_{\text{later}} = n\Delta t + M_{\text{now}} \quad v_{\text{esc}} = \sqrt{\frac{2\mu}{r}}$$

$$M = E - \sin E \quad M = e \sinh H - H \quad T^2 = 4\pi^2 a^3 / \mu$$

$$\frac{v^2}{2} - \frac{\mu}{r} = \varepsilon \quad \varepsilon = -\frac{\mu}{2a} \quad G \approx 6.67 \times 10^{-11} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2}$$

2-Body: $\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r}$ N-Body: $\ddot{\mathbf{R}}_i = G \sum_{j=1, j \neq i}^N \frac{m_j}{r_{ij}^3} \mathbf{r}_{ij}$

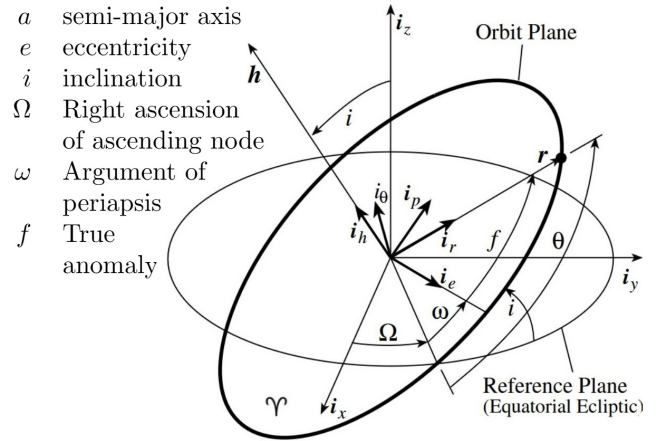
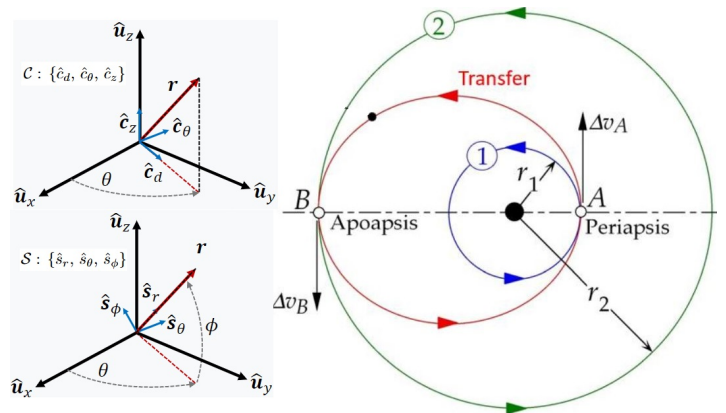
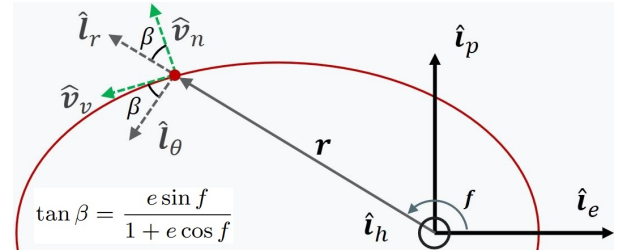
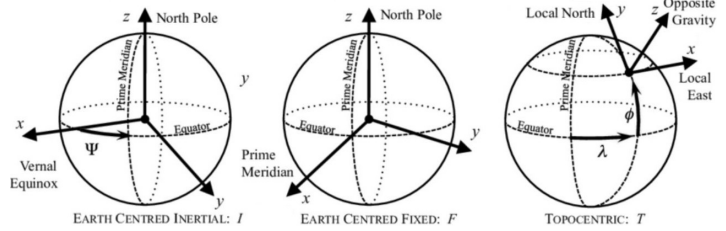
$r_{\text{SOI of } m_2} = r_{21} \left(\frac{m_2}{m_3} \right)^{2/5}$ For $m_3 > m_2 \gg m_1$

For hyperbolics $\begin{cases} x = a(e - \cosh H) & f_{\infty} = \pm a \cos -e^{-1} \\ y = a\sqrt{e^2 - 1} \sinh H & \delta = \pm 2a \sin e^{-1} \end{cases}$

$I_{sp} = \frac{v_e}{g_0} = \frac{T}{\dot{m}g_0}$ Rocket eq.: $\Delta v = v_e \ln \left(\frac{m_0}{m_f} \right)$

Hohmann transfer: $\Delta v_A = |v_{T,p} - v_1|$, $\Delta v_B = |v_2 - v_{T,a}|$

- Cylindrical**, $\mathcal{C}: \{\hat{c}_d, \hat{c}_\theta, \hat{c}_z\}$, $\mathbf{C}\mathbf{r} = d\hat{c}_d + z\hat{c}_z$,
 $\theta = \arccos \left(\frac{\hat{u}_x \cdot \mathbf{r}_{xy}}{d} \right)$, ${}_{\mathcal{U}}^{\mathcal{C}}\mathbf{R} = \mathbf{R}_z(\theta)$
- Spherical**, $\mathcal{S}: \{\hat{s}_r, \hat{s}_\theta, \hat{s}_\phi\}$, $\mathbf{S}\mathbf{r} = r\hat{s}_r$, $\theta = \arccos \left(\frac{\hat{u}_x \cdot \mathbf{r}_{xy}}{\|\mathbf{r}_{xy}\|} \right)$,
 $\phi = \arccos \left(\frac{\hat{u}_z \cdot \mathbf{r}_z}{r} \right)$, ${}_{\mathcal{U}}^{\mathcal{S}}\mathbf{R} = \mathbf{R}_y(\phi)\mathbf{R}_z(\theta)$
- Earth-Centered Inertial (ECI)**, $\mathcal{I}$
 $\{\hat{i}_x, \hat{i}_y, \hat{i}_z\} = \{\text{First point of Ares}, \hat{i}_z \times \hat{i}_x, \text{True North pole}\}$
- Earth-Centered Earth-Fixed (ECEF)**, $\mathcal{E}$
 $\{\hat{e}_x, \hat{e}_y, \hat{e}_z\} = \{\text{Greenwich meridian}, \hat{e}_z \times \hat{e}_x, \text{True North Pole}\}$
 $\text{LST} = \text{GMST} + \text{Long.} \iff \gamma(t) = \gamma_G(t) + \lambda \quad {}_{\mathcal{I}}^{\mathcal{E}}\mathbf{R} = \mathbf{R}_z(\gamma_G(t))$
- Topocentric** (some combo of up, east, north), $\mathcal{T}$
- Perifocal** $\mathcal{P}: \{\hat{i}_e, \hat{i}_p, \hat{i}_h\} = \{\text{Ecc.}, \text{Parameter}, \text{out-of-plane}\}$
Motion confined to plane, so $\mathbf{r} = x\hat{i}_e + y\hat{i}_p = r \cos f \hat{i}_e + r \sin f \hat{i}_p$
- Orbit**, $\mathcal{O}: \{\hat{i}_r, \hat{i}_\theta, \hat{i}_h\}$, **Velocity**, $\mathcal{V}: \{\hat{v}_r, \hat{v}_n, \hat{v}_h\}$, ${}_{\mathcal{O}}^{\mathcal{V}}\mathbf{R} = \mathbf{R}_z(\beta)$
 $\mathbf{h} = \mathbf{r} \times \mathbf{v} = r^2 \dot{f} \hat{i}_h \quad \boldsymbol{\omega}_{\mathcal{O}/\mathcal{P}} = \dot{f} \hat{i}_h \quad \mathbf{v} = (he \sin f)/p \hat{v}_r + h/r \hat{v}_\theta$
- 3D Orientation** $\alpha = [a \ e \ i \ \Omega \ \omega \ (f \text{ XOR } M_0)]^T$
 ${}_{\mathcal{I}}^{\mathcal{P}}\mathbf{R} = \mathbf{R}_z(\omega)\mathbf{R}_x(i)\mathbf{R}_z(\Omega) = [\hat{i}_e \ \hat{i}_p \ \hat{i}_h] \quad i = \arccos({}_{\mathcal{I}}^{\mathcal{P}}\mathbf{R}[3, 3])$
 $\Omega = \text{atan} \left(\frac{{}_{\mathcal{I}}^{\mathcal{P}}\mathbf{R}[3, 1]}{-{}_{\mathcal{I}}^{\mathcal{P}}\mathbf{R}[3, 2]} \right) \quad \omega = \text{atan} \left(\frac{{}_{\mathcal{I}}^{\mathcal{P}}\mathbf{R}[1, 3]}{{}_{\mathcal{I}}^{\mathcal{P}}\mathbf{R}[2, 3]} \right)$



Quantity	Circular	Elliptical	Parabolic	Hyperbolic
Parametric equation	$x^2 + y^2 = r^2$	$x^2/a^2 + y^2/b^2 = 1$	$x^2 = 2py$	$x^2/a^2 - y^2/b^2 = 1$
Eccentricity	$e = 0$	$e = \sqrt{a^2 - b^2}/a \quad 0 < e < 1$	$e = 1$	$e = \sqrt{a^2 + b^2}/a \quad e > 1$
Perifocal distance	$q = a$	$q = a(1 - e)$	$q = p/2$	$q = a(1 - e)$
Mean angular motion	$n = \sqrt{\mu/a^3}$	$n = \sqrt{\mu/a^3}$	$n = \sqrt{\mu}$	$n = \sqrt{\mu/a^3}$
Period	$T = 2\pi/n$	$T = 2\pi/n$	$T = \infty$	$T = \infty$
Anomaly	$f = M = E$	$\tan \left(\frac{f}{2} \right) = \sqrt{\frac{1+e}{1-e}} \tan \left(\frac{E}{2} \right)$	$\tan \left(\frac{f}{2} \right) = \frac{D}{\sqrt{2q}}$	$\tan \left(\frac{f}{2} \right) = \sqrt{\frac{e+1}{e-1}} \tanh \left(\frac{H}{2} \right)$
Mean anomaly	$M = M_0 + nt$	$M = E - e \sin E$	$M = qD + D^3/6$	$M = e \sinh H - H$
Radius (from focus)	$r = a$	$r = a(1 - e \cos E)$	$r = q + D^2/2$	$r = a(1 - e \cosh H)$
Range rate	$r\dot{r} = 0$	$r\dot{r} = e\sqrt{a\mu} \sin E$	$r\dot{r} = \sqrt{\mu}D$	$r\dot{r} = e\sqrt{a\mu} \sinh H$
Specific total orbit energy	$\varepsilon < 0$	$\varepsilon < 0$	$\varepsilon = 0$	$\varepsilon > 0$

Conversions

- Cartesian to Classical

1. $v_r = \mathbf{v} \cdot \hat{\mathbf{r}}, v_\perp^2 = v^2 - v_r^2$
2. $\mathbf{h} = \mathbf{r} \times \mathbf{v}$
3. $i = \arccos(\mathbf{h}_z/h), \mathbf{N} = \hat{\mathbf{k}} \times \mathbf{h}$
4. $\Omega = \arccos(\mathbf{N}_x/N)$ if $\mathbf{N}_y \geq 0$ else $2\pi - \arccos(\mathbf{N}_x/N)$
5. $\mathbf{e} = (\mathbf{v} \times \mathbf{h})/\mu$
6. $p = h^2/\mu, a = p/(1 - e^2)$
7. $\omega = \arccos(\frac{\mathbf{e} \cdot \mathbf{N}}{eN})$ if $\mathbf{e}_z \geq 0$ else $2\pi - \arccos(\frac{\mathbf{e} \cdot \mathbf{N}}{eN})$
8. $f = \arccos(\frac{\mathbf{e} \cdot \mathbf{r}}{er})$ if $v_r \geq 0$ else $2\pi - \arccos(\frac{\mathbf{e} \cdot \mathbf{r}}{er})$

Note: This should be updated to instead use $\mathcal{P}_{\mathcal{I}}\mathbf{R}$ method for extracting angles.

- Classical to Cartesian

1. $p = a(1 - e^2), r = p/(1 + e \cos f), h = \sqrt{\mu p}$
2. $\dot{f} = h/r^2, \dot{r} = (he \sin f)/p$
3. Assemble state in Perifocal, and rotate to Inertial

$$\mathcal{P}\mathbf{x} = \begin{bmatrix} r \cos f \\ r \sin f \\ 0 \\ -r\dot{f} \sin f + \dot{r} \cos f \\ r\dot{f} \cos f + \dot{r} \sin f \\ 0 \end{bmatrix} \quad \mathcal{I}\mathbf{x} = \begin{bmatrix} \mathcal{I}_{\mathcal{P}}\mathbf{R} & \mathbf{0} \\ \mathbf{0} & \mathcal{I}_{\mathcal{P}}\mathbf{R} \end{bmatrix} \mathcal{P}\mathbf{x}$$

Kepler's Laws

1. The orbit of a planet is an ellipse with the Sun at one of the two foci.
2. A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time.
3. The square of a planet's orbital period is proportional to the cube of the length of the semi-major axis of its orbit $T^2 = 4\pi^2 a^3/\mu$.

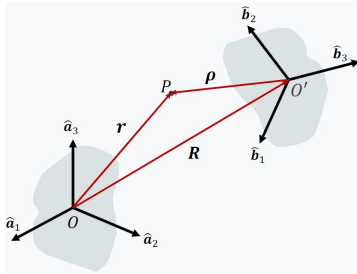


Figure 1: Two frames in motion. Source: Lecture 1, slide 17.

In relation to the frames depicted in Fig. 1, and beginning from the following expression for the velocity of point P

$$(\mathbf{v}_P)_{\mathcal{A}} = (\mathbf{v}_{O'})_{\mathcal{A}} + (\mathbf{v}_P)_{\mathcal{B}} + \boldsymbol{\omega}_{\mathcal{B}/\mathcal{A}} \times \boldsymbol{\rho}$$

the acceleration of point P , $(\mathbf{a}_P)_{\mathcal{A}}$, is found by application of the Transport Theorem, presented here by example

$$\text{Transport Theorem:} \quad \frac{d}{dt}(\mathcal{E}\mathbf{r}_{\mathcal{N}}) = \frac{d}{dt}(\mathcal{E}\mathbf{r}_{\mathcal{E}}) + \boldsymbol{\omega}_{\mathcal{E}/\mathcal{N}} \times \mathcal{E}\mathbf{r}$$

and is

$$\begin{aligned} (\mathbf{a}_P)_{\mathcal{A}} &= \frac{d}{dt}(\mathcal{A}\mathbf{v}_{O'})_{\mathcal{A}} + \frac{d}{dt}(\mathcal{B}\mathbf{v}_P)_{\mathcal{B}} + \boldsymbol{\omega}_{\mathcal{B}/\mathcal{A}} \times \mathcal{B}\mathbf{v}_P + \frac{d}{dt}\boldsymbol{\omega}_{\mathcal{B}/\mathcal{A}} \times \boldsymbol{\rho} + \boldsymbol{\omega}_{\mathcal{B}/\mathcal{A}} \times \frac{d}{dt}\boldsymbol{\rho} \\ &= (\mathcal{A}\mathbf{a}_{O'})_{\mathcal{A}} + (\mathcal{B}\mathbf{a}_P)_{\mathcal{B}} + \boldsymbol{\omega}_{\mathcal{B}/\mathcal{A}} \times \mathcal{B}\mathbf{v}_P + \boldsymbol{\alpha}_{\mathcal{B}/\mathcal{A}} \times \boldsymbol{\rho} + \boldsymbol{\omega}_{\mathcal{B}/\mathcal{A}} \times ((\mathcal{B}\mathbf{v}_P) + \boldsymbol{\omega}_{\mathcal{B}/\mathcal{A}} \times \boldsymbol{\rho}) \\ &= \boxed{(\mathcal{A}\mathbf{a}_{O'})_{\mathcal{A}} + (\mathcal{B}\mathbf{a}_P)_{\mathcal{B}} + 2\boldsymbol{\omega}_{\mathcal{B}/\mathcal{A}} \times \mathcal{B}\mathbf{v}_P + \boldsymbol{\alpha}_{\mathcal{B}/\mathcal{A}} \times \boldsymbol{\rho} + \boldsymbol{\omega}_{\mathcal{B}/\mathcal{A}} \times \boldsymbol{\omega}_{\mathcal{B}/\mathcal{A}} \times \boldsymbol{\rho} = (\mathbf{a}_P)_{\mathcal{A}}} \end{aligned}$$

with the following names for each term:

- Linear acceleration of frame O' : $(\mathcal{A}\mathbf{a}_{O'})_{\mathcal{A}}$
- Linear acceleration of point P in frame $\mathcal{B}$: $(\mathcal{B}\mathbf{a}_P)_{\mathcal{B}}$
- Coriolis acceleration: $2\boldsymbol{\omega}_{\mathcal{B}/\mathcal{A}} \times \mathcal{B}\mathbf{v}_P$
- Euler acceleration: $\boldsymbol{\alpha}_{\mathcal{B}/\mathcal{A}} \times \boldsymbol{\rho}$
- Centrifugal acceleration: $\boldsymbol{\omega}_{\mathcal{B}/\mathcal{A}} \times \boldsymbol{\omega}_{\mathcal{B}/\mathcal{A}} \times \boldsymbol{\rho}$